

FINAL YEAR PROJECT — PFE

ESPRIT · 2026

PEAXIS

An AI-Powered Hiring Operating System

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● Software Engineering Intern · Full-Stack & AI Systems Architect

Next.js

NestJS

FastAPI

PostgreSQL

Gemini

Docker

Prospecter — Internship Host Company

An AI-powered B2B sales prospecting platform — automating outbound pipeline generation for growth-stage companies through autonomous AI SDR technology.

MISSION

Replace manual sales prospecting with a fully autonomous AI SDR — enabling B2B companies to scale outbound revenue generation without scaling headcount.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine

Autonomous AI sales development rep — qualifies leads, drafts outreach, and orchestrates follow-up sequences at scale.



Lead Intelligence

Enriches prospect data with firmographics, intent signals, and behavioural triggers to prioritise highest-value targets.



Outbound Automation

Multi-channel sequences (email, LinkedIn) with LLM personalisation — zero manual copy-paste prospecting.



Pipeline Analytics

Real-time conversion tracking, A/B testing for messaging, and CRM-sync to close the loop from prospect to revenue.

 **Strategic context** — Although Prospecter specialises in AI-powered sales prospecting, its product and engineering environment — LLM pipelines, multi-tenant SaaS, background workers, Redis caching — provided the architectural foundation for developing PEAXIS as a next-generation HR technology platform during this internship.

From AI Sales Infrastructure to AI Hiring Infrastructure

A 6-month final year internship at Prospector — sole engineer responsible for design, architecture, and full implementation of PEAXIS.

PROSPECTER ENVIRONMENT

- LLM pipeline architecture
- Multi-tenant SaaS patterns
- Redis caching strategies
- AI-as-a-service design



PEAXIS INNOVATION

- AI-powered CV parsing & scoring
- Explainable hiring decisions
- Async background workers
- Candidate-facing portal UX



Dec 22

Kickoff · Prospector



Jan

Architecture design



Feb

Core API + Auth



Mar

AI service + pipeline



Apr

Frontends + billing



May

Polish + defense prep



Full-Stack Architecture

- NestJS API (14 modules)
- Next.js Hire + Jobs portals
- Multi-tenant data model
- BullMQ background pipeline



AI Integration

- FastAPI AI microservice
- CV parsing (Gemini)
- Match scoring engine
- Hiring Copilot · JD Generator



Platform Engineering

- Docker dev + prod setup
- Prisma schema + migrations
- pgvector semantic search
- Stripe billing integration

SOFTWARE ENGINEERING INTERN

AI INTEGRATION

PLATFORM ARCHITECT

SOLO CONTRIBUTOR

3 THE PROBLEM

Recruitment is **broken** at every layer

The hiring process hasn't fundamentally changed in 20 years. Time, money, and talent are wasted at every step.

42 days

Average time-to-hire

SHRM 2024

75%

Recruiters overwhelmed by CV volume

LinkedIn Talent Report

60%

Candidates ghost after poor UX

Indeed Survey 2023



Screening bottleneck

Recruiters spend 23 hours per week manually reviewing CVs. 80% of that time is wasted on unqualified applications.



Bias & inconsistency

Manual screening introduces unconscious bias and inconsistent evaluation criteria across reviewers and hiring rounds.



Candidate friction

Long applications, no feedback, and opaque processes cause top talent to abandon their candidacies before completion.

Global impact: companies lose an average of **\$14,900** per bad hire, while top candidates accept competing offers in under 10 days.

Regional context: In Tunisia & North Africa, growing SMEs rely on spreadsheets and informal networks — modern ATS adoption remains critically low among early-stage companies.

3 THE PROBLEM

Manual workflows, legacy ATS & fragmented stacks are **not enough**

From spreadsheets to traditional ATS to stitched-together SaaS tools — none deliver the AI-native intelligence modern hiring requires.

CAPABILITY	MANUAL	LEGACY ATS	FRAGMENTED	PEAXIS
AI-powered candidate ranking	✗	✗	partial	✓
Explainable match scores	✗	✗	✗	✓
Automatic CV parsing	✗	✗	partial	✓
Candidate-facing portal	✗	basic	basic	✓
Real-time analytics pipeline	✗	✗	✗	✓
Semantic job search	✗	✗	✗	✓
Hiring copilot / AI assistant	✗	✗	✗	✓
Multi-tenant SaaS architecture	✗	partial	✗	✓

3 THE PROBLEM

The market gap: intelligent + integrated + explainable



AI black boxes

"This candidate scored 72" — but why? Recruiters cannot act on scores they cannot explain. Bias complaints follow.



Fragmented tooling

Notion for job specs. Greenhouse for pipeline. Calendly for interviews. Zero integration — context lost at every handoff.



No candidate intelligence

Existing tools track applications. None actively guide recruiters on who to advance, how to interview, or what risks to flag.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



PEAXIS fills this exact gap

AI scores are explainable. Tools are integrated. The candidate experience is first-class. That's the OS layer missing from the market.

4 FUNCTIONAL REQUIREMENTS

Core platform requirements

FR-01 through FR-05 cover the foundational platform capabilities required for a production-grade SaaS recruitment system.



FR-01

Authentication & Security

- Email/password with Argon2 hashing
- OAuth2: Google + LinkedIn
- JWT access (15m) + refresh (7d)
- Rate limiting: 1000 req/60s



FR-02

Multi-Tenant Architecture

- Business-scoped data isolation
- X-Business-ID header enforcement
- User ↔ Business join table
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise tiers
- JwtAuthGuard + PlanGuard
- Business member roles



FR-04

Job Management

- Full CRUD with rich filters
- Semantic search via pgvector
- Auto embedding generation
- Type, location, salary, skills filters



FR-05

Application Pipeline

- 6-stage:
Applied → Screening → Interview → Offer → Hired/Rejected
- Stage transition history
- AI scoring at application time
- Recruiter kanban view

4 FUNCTIONAL REQUIREMENTS

AI & analytics requirements

FR-06 through FR-10 define the intelligence layer — the core differentiator that elevates PEAXIS above traditional ATS tools.



FR-06

AI Match Scoring

- Deterministic skill-overlap engine
- Score: 0–100 with explanation
- Strengths & weaknesses breakdown
- Applied at submission time



FR-07

CV Parsing

- PDF / DOCX / TXT support
- Gemini 2.5–flash extraction
- Structured JSON output
- Auto pre-fills candidate profile



FR-08

Candidate Onboarding

- Anonymous apply-with-CV flow
- Profile completeness scoring
- Semantic embedding generation
- Email verification



FR-09

Interview Scheduling

- Interview CRUD with type flags
- Linked to application record
- Interviewer assignment
- Status tracking



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking
- Per-job + global analytics

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods behind load balancer. BullMQ handles surge traffic via Redis queue. pgvector scales to millions of embeddings.



OWASP aligned

NFR-02 **Security**

Argon2 passwords, JWT rotation, OAuth2, X-Business-ID multi-tenant isolation, OWASP-hardened API, X-Service-Secret inter-service auth.



p95 < 200ms

NFR-03 **Performance**

Redis 3-layer caching (embeddings 7d, CV parse 30d, AI results 24h). Sub-200ms API p95. Async BullMQ for heavy operations.



99.5% target SLA

NFR-04 **Availability**

Preferred-region retries in SDK. BullMQ retry logic. Docker health checks. Graceful degradation on AI service downtime.



Full audit trail

NFR-05 **Explainability**

Every AI score exposes: numeric value, label (Strong/Good/Weak), matched skills, missing skills, and a plain-language explanation.



Microservices

NFR-06 **Modularity**

NestJS feature modules. Independent AI service. Decoupled frontend apps. BullMQ workers isolated from request handlers.



Cache-first AI

NFR-07 **Cost Optimization**

Gemini Flash for fast tasks. GPT-4.1-mini for quality outputs. Redis caching prevents redundant LLM calls. Docker resource limits.

PEAXIS — Integrated AI Hiring Operating System

Four deeply integrated products that replace the fragmented stack every recruiter tolerates today — built on SaaS architecture principles refined at Prospector.



API Layer
PEAXIS Core

NestJS REST API — single source of truth. Auth, billing, multi-tenant, all data owned here.



Recruiter Dashboard
PEAXIS Hire

Next.js SPA — pipeline Kanban, AI insights panel, job management, analytics.



Candidate Portal
PEAXIS Jobs

Next.js SPA — semantic job search, CV upload & parse, application tracking.



FastAPI Service
AI Brain

Stateless Python service — CV parsing, match scoring, copilot, JD generation.

● All four products share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

PEAXIS Hire — Recruiter Dashboard

NEXT.JS 16

The workspace where recruiters live. Every hiring decision powered by AI, every action one click away.



Pipeline Kanban

Drag-and-drop stage management with AI scores visible on every card



AI Insight Panel

Match score, matched/missing skills, AI explanation per candidate



Job Analytics

Funnel metrics, score distributions, time-to-stage per job



Inline Editing

Edit job details inline without leaving the workspace — no navigation

The screenshot displays the PEAXIS Hire Recruiter Dashboard for a job titled "Senior React Developer". The browser address bar shows "hire.peaxis.com/jobs/senior-react-dev". The dashboard includes a header with the PEAXIS logo, the job title, and a "Published" status. Below the header, there are tabs for Pipeline, Candidates, Analytics, Activity, and Job Details. The Pipeline tab is active, showing a Kanban board with stages: Applied (12), Screening (7), Interview (4), Offer (2), and Hired (1). Each stage has a corresponding candidate card with a match score (e.g., 91% for Applied, 78% for Screening). On the right side, there is a sidebar for a specific candidate, Sara Mansouri, showing her status as "Applied" (3 days ago) and an "AI Analysis" section powered by PEAXIS AI. At the bottom of the sidebar, there are buttons for "Move to Screening", "Reject", and "View Full Profile".

Architecture: Server components fetch all data in parallel (Promise.all). Client components own interactive state only. Candidate card click opens a slide-in **CandidateDrawer** — never navigates away.

PEAXIS Jobs — Candidate Portal

NEXT.JS 16

The candidate experience that makes applying frictionless — from discovery to tracking.

The screenshot shows a web browser window with the URL `jobs.peaxis.com`. The page features a header with the PEAXIS logo, navigation links for Jobs, Applications, and Profile, and a user profile button for Sara M. Below the header is a search bar with the placeholder text "Search roles, companies, skills..." and a Search button. A filters section includes buttons for Tunisia, Remote OK, Full-time, and AI-scored. The main content area displays three job listings:

- Senior React Developer** (TechCorp Tunisia · Tunis, TN) with a 91% Strong rating, 2 days ago, and an Apply button. Skills: React, TypeScript, Node.js. Salary: 3,500–4,500 TND.
- Python AI Engineer** (AISolutions · Remote) with an 84% Good rating, 5 days ago, and an Apply button. Skills: Python, FastAPI, LLM. Salary: 4,000–5,500 TND.
- Product Designer** (DesignHub · Hybrid · Sfax) with a 72% Good rating, 1 week ago, and an Apply button. Skills: Figma, UX, Mobile. Salary: 2,800–3,800 TND.

At the bottom, there is a section for "Upload your CV" with the text "PEAXIS AI extracts your profile automatically" and an Upload CV button.



Semantic Job Search

pgvector-powered search — finds relevant jobs even with different wording



Apply with CV

Upload CV → AI parses it → profile pre-filled → one-click apply



Application Tracker

Real-time pipeline stage visibility — Applied, Screening, Interview, Offer

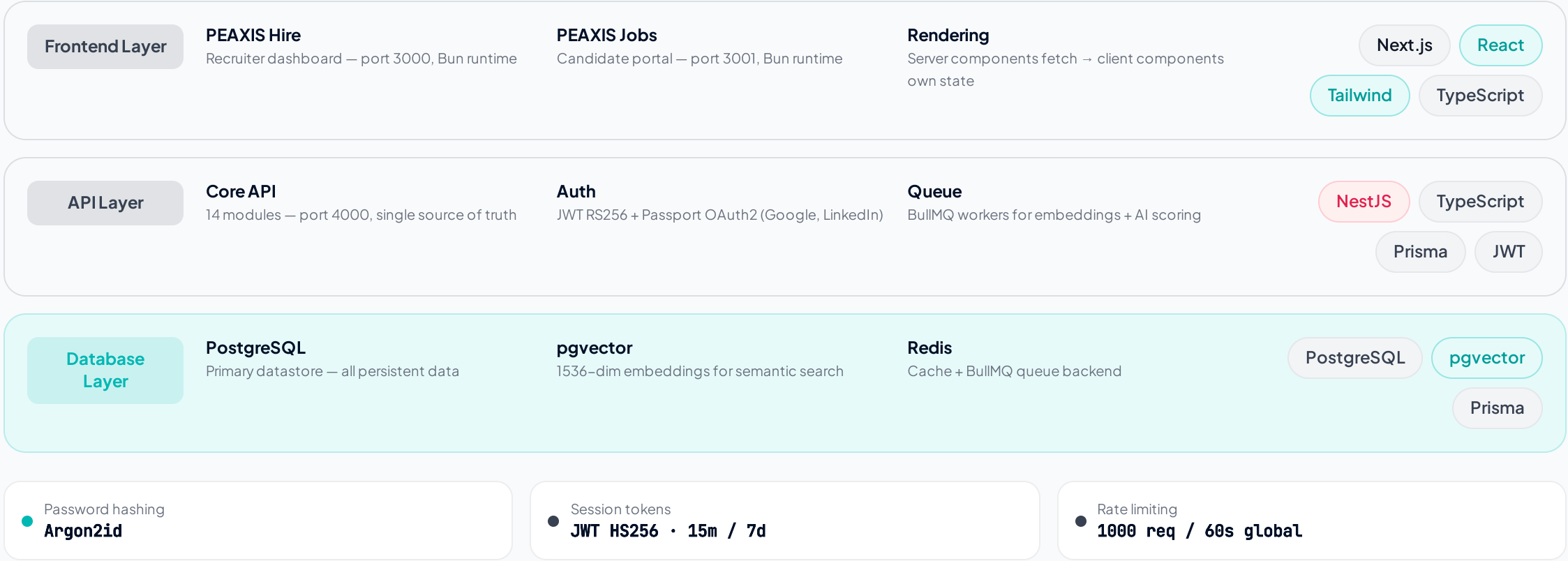


Job Recommendations

Personalised job feed powered by candidate embedding similarity

Upload CV (PDF/DOCX) → AI parses (Gemini) → Profile pre-filled → One-click apply → Real-time status

Frontend & API stack



AI & infrastructure stack

AI FEATURE	MODEL	WHY
● CV Parsing	Gemini 2.5-flash	Fast structured JSON, low cost
● Match Scoring	Deterministic	Reproducible, explainable, no LLM needed
● JD Generation	Gemini 2.5-flash	Structured output + bias checking
● Hiring Copilot	Gemini 2.5-flash	Summaries, risk flags, interview Qs
● Cover Letters	GPT-4.1-mini	Higher creative quality
● Embeddings	text-embedding-3-small 1536-dim, cost-effective	

Infrastructure

FastAPI

AI service — stateless, Python 3.12+, Pydantic schemas

Redis

3-layer cache: embeddings 7d · CV parse 30d · results 24h

BullMQ

Background queue for embeddings, scoring, analysis jobs

Docker

Dev + prod compose files, multi-stage Dockerfiles

Stripe

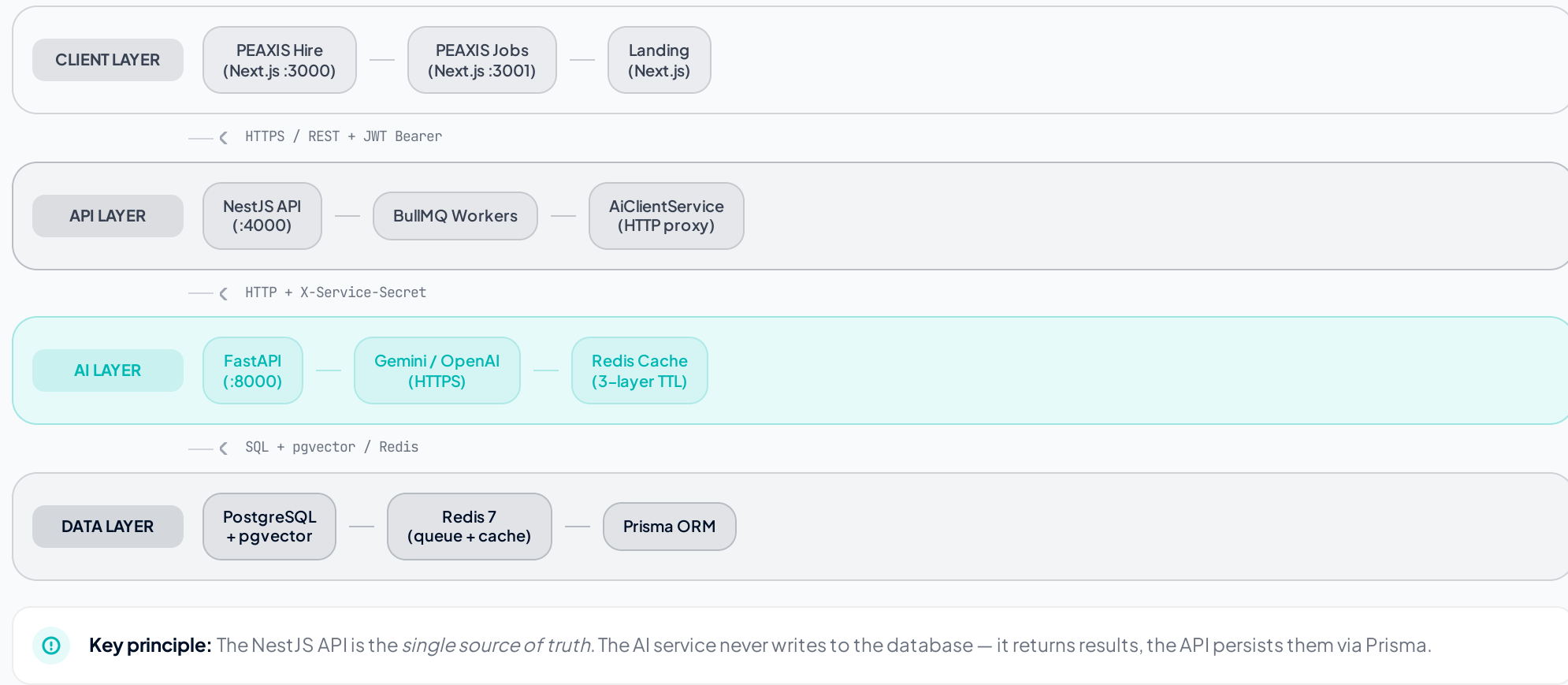
Subscription billing with webhook event handling

Redis Cache-first strategy: sha256(model+input) key → hit avoids LLM call entirely. Embeddings cached 7 days, CV parses 30 days, general AI results 24 hours.

8 LOGICAL ARCHITECTURE

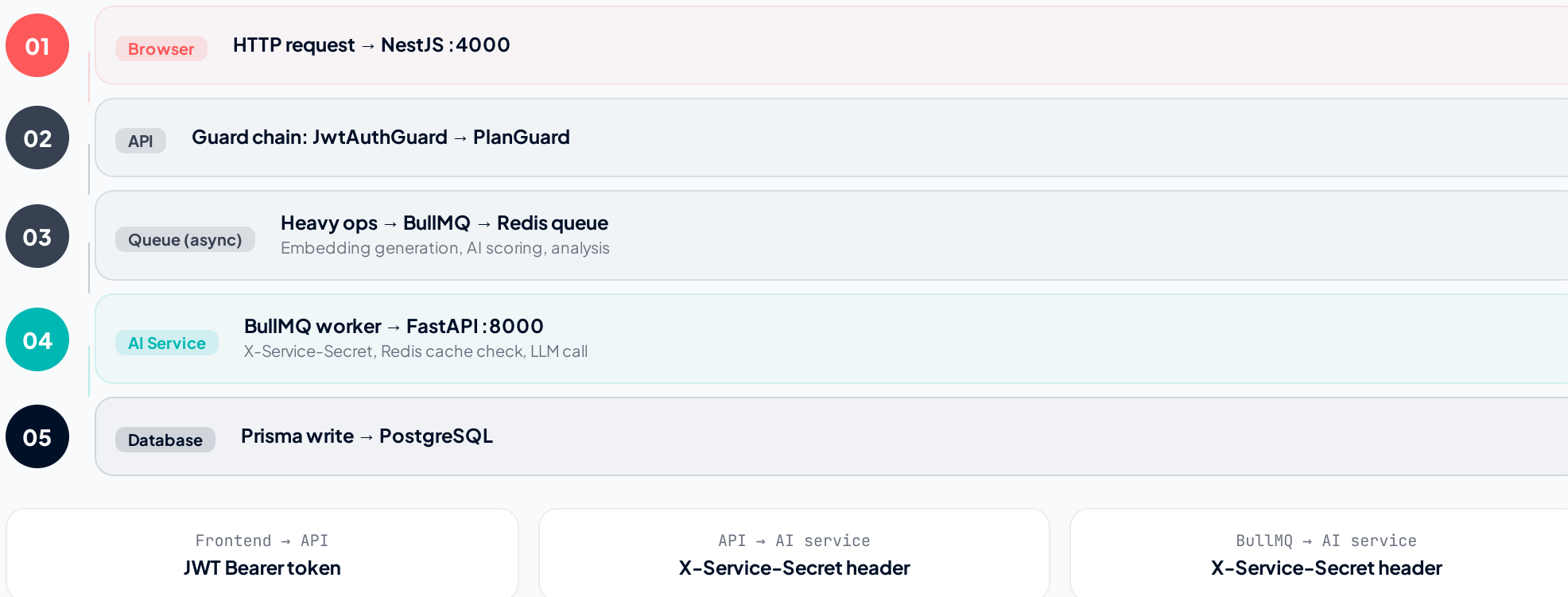
System layered architecture

Each layer has a single, clear responsibility. No layer bypasses the one above it.

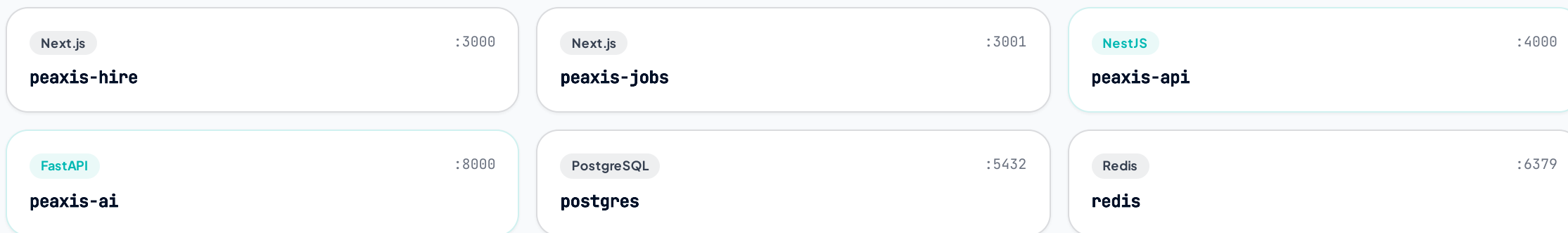


Request lifecycle & data flow

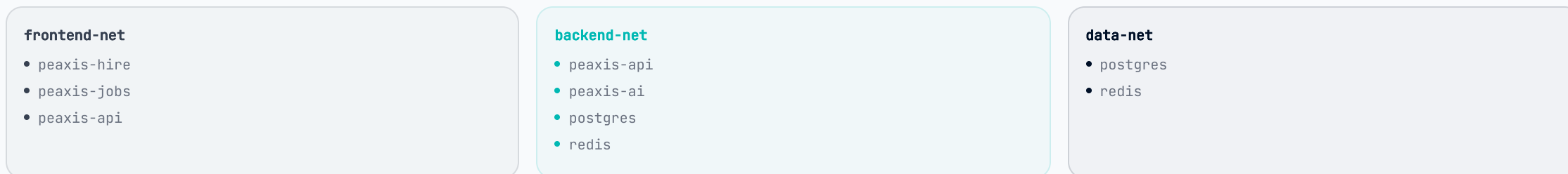
From browser to database — every hop is authenticated, scoped, and auditable.



Docker Compose deployment topology



DOCKER NETWORKS



Exposed to host: :3000 (Hire) :3001 (Jobs) :4000 (API) :8000 (AI) :5432 (PG dev) :6379 (Redis dev)

Infrastructure & DevOps setup

● Development

`docker-compose.yml`

Infra only (Postgres, Redis). Frontend/API run natively with hot-reload.

- postgres:5432
- redis:6379
- API: native 4000
- AI: native 8000

● Production

`docker-compose.prod.yml`

All services containerised with multi-stage Dockerfiles and resource limits.

- All 6 services
- Multi-stage builds
- Resource limits
- Health checks

🔒 Secrets management

All credentials via environment variables. `.env.example` committed. No secrets in version control.

⚙️ Config via ConfigService

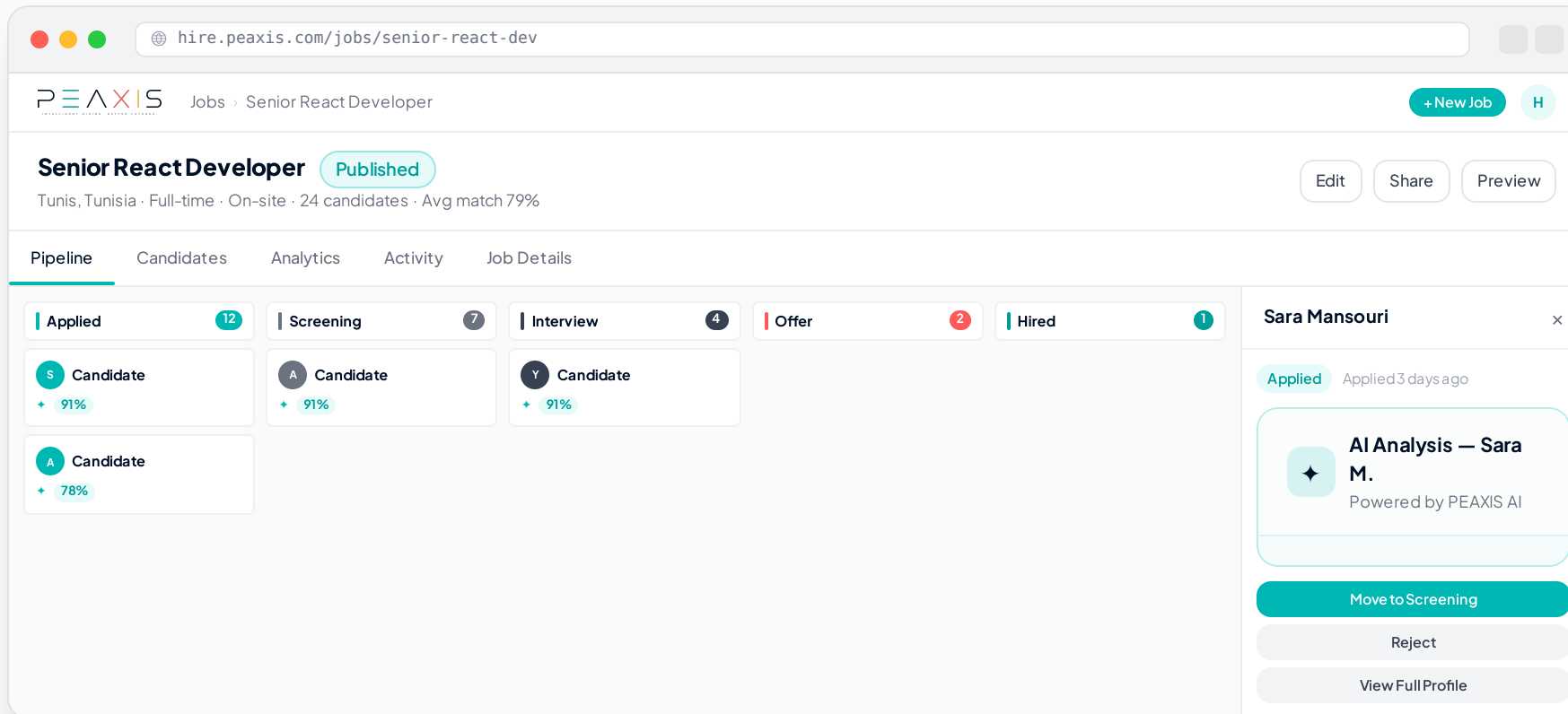
NestJS ConfigService wraps `process.env`. Direct `process.env` access is prohibited in application code.

↻ Prisma migrations

Schema-first with `prisma migrate dev`. All migrations version-controlled in `prisma/migrations/`.

Observability: NestJS Logger / HumanysLogger for structured logs. No `console.log`. FastAPI logs routed via `uvicorn`. Redis diagnostic string captured on latency spikes.

Demo: Recruiter Dashboard



Job Workspace

All job-related data in one tabbed view — no page hops

Pipeline Kanban

Drag-and-drop stage management with AI scores on every card

Candidate Drawer

Slide-in panel reveals AI insights without leaving the pipeline

Inline Job Edit

Edit job details within the workspace — `router.refresh()` on save

UX principle: Candidate card click *never navigates* — it opens the CandidateDrawer (slide-in panel). "View Full Profile" inside the drawer navigates. One action, clear intent.

Demo: Candidate Flow



jobs.peaxis.com

PEAXIS

Jobs Applications Profile

Sara M.

Q Search roles, companies, skills...

Search

Filters:

Tunisia

Remote OK

Full-time

AI-scored

T

Senior React Developer

TechCorp Tunisia · Tunis, TN

React TypeScript Node.js 3,500–4,500 TND

★ 91% Strong

2 days ago

Apply →

A

Python AI Engineer

AI Solutions · Remote

Python FastAPI LLM 4,000–5,500 TND

★ 84% Good

5 days ago

Apply →

D

Product Designer

DesignHub · Hybrid · Sfax

Figma UX Mobile 2,800–3,800 TND

★ 72% Good

1 week ago

Apply →

↑

Upload your CV

PEAXIS AI extracts your profile automatically

Upload CV

PIPELINE

PDF / DOCX → Gemini 2.5–flash → JSON Profile → Redis Cache (30d) → Score + Apply → Pipeline Stage

Upload CV once — AI handles the rest.

Demo: AI Intelligence Layer

● Match Scoring

Deterministic skill-overlap → 0–100 with matched/missing skills + plain-language explanation

● CV Parsing

Gemini 2.5–flash → structured JSON profile from PDF/DOCX in under 3 seconds

● Hiring Copilot

Candidate summary · interview questions · risk flags · next action — all in-context

● JD Generator

Bias-checked job description from title + bullets, with skill requirement suggestions

★ AI Decision Engine

INPUT

CV (PDF)

+

Job Description

ENGINE

Parse CV

→ Embed Skills

→ pgvector Match

→ Score + Explain

OUTPUT

82 /100

Strong Match

React ✓

TypeScript ✓

Node.js ✓

PostgreSQL ✓

GraphQL ✗

Docker ✗

"Strong React + TS background. GraphQL gap is trainable given existing Node.js depth."

Design rule: Every AI score exposes — numeric value · confidence label · matched skills (teal) · missing skills (coral) · plain-language explanation. No silent black boxes.

Technical achievements & operational outcomes

14

NestJS Feature Modules

auth, jobs, apps, AI, billing...

60+

REST API Endpoints

documented with Swagger

6

AI Features

parse · score · copilot · JD ·
cover letter · embedding

4

Microservices

Hire · Jobs · API · AI

3

Redis Cache Layers

embeddings 7d · CV 30d ·
results 24h

100%

Multi-tenant Isolation

every record scoped to
businessId

NOTEWORTHY IMPLEMENTATIONS

pgvector 1536-dim semantic search

Playwright PDF export pipeline

OAuth2: Google + LinkedIn

Stripe webhook billing

BullMQ background processor

Prisma schema-first migrations

Argon2 password hashing

Docker multi-stage builds

TypeScript

65%

Python

20%

SQL/Prisma

10%

CSS/Tailwind

5%

OPERATIONAL OUTCOMES

Reduced screening friction

Automated CV parsing & AI ranking eliminates
manual shortlisting

Faster candidate processing

Async BullMQ workers process scoring without
blocking user actions

Explainable AI decisions

Every match score includes matched skills, gaps &
natural language rationale

Improved candidate UX

Semantic search, real-time status tracking &
transparent pipeline feedback

Product roadmap

Phase 1 — Core Platform (Done)

- ✓ NestJS API with 14 modules
- ✓ Multi-tenant auth + OAuth2
- ✓ Application pipeline (6 stages)
- ✓ Job CRUD + semantic search
- ✓ Stripe billing integration

Phase 2 — AI Layer (Done)

- ✓ FastAPI AI microservice
- ✓ CV parsing + match scoring
- ✓ Hiring Copilot + JD Generator
- ✓ BullMQ embedding pipeline
- ✓ Redis 3-layer caching

Phase 3 — Frontends (Done)

- ✓ PEAXIS Hire recruiter dashboard
- ✓ PEAXIS Jobs candidate portal
- ✓ Job Workspace with tabs + drawer
- ✓ Analytics dashboard
- ✓ Interview scheduling

Phase 4 — Near-term (Realistic)

- Email notification system
- GDPR compliance toolkit
- API rate limiting per plan
- End-to-end Playwright tests
- Candidate referral engine

Phase 5 — Long-term (Visionary)

- Video interview + AI scoring
- Mobile-first candidate app
- Predictive attrition modeling
- Multi-region deployment
- Enterprise SSO + audit logs

☒ Done ☐ In Progress ☐ Planned

Challenges & lessons learned

01

CHALLENGE

Multi-tenant data isolation at scale

Every query, guard, and background job must be scoped to a businessId without leaking data between tenants.

RESOLUTION

X-Business-ID header enforced at controller level. PrismaService always receives businessId from the guard. BullMQ jobs include businessId in payload.

KEY LESSON

Design security boundaries first — retrofitting scoping is 10x harder.

02

CHALLENGE

AI explainability vs. LLM cost

Deterministic scoring is cheap but limited. LLM scoring is rich but expensive and non-reproducible at scale.

RESOLUTION

Hybrid approach: deterministic skill-overlap for the score (reproducible, zero cost), Gemini Flash for the explanation text (low cost, high quality, cached 24h).

KEY LESSON

Separate "what" (deterministic) from "why" (LLM). Cache aggressively.

03

CHALLENGE

Frontend state vs. server freshness

Next.js server components cache aggressively. After inline edits (job form, stage moves), UI must reflect the updated state without full navigation.

RESOLUTION

Server actions call router.refresh() after mutations — triggers server component re-fetch without navigation. Client components own transient state only.

KEY LESSON

Understand Next.js cache invalidation deeply before building complex workspaces.

Future scope & roadmap



Video Interview Integration

Embedded video screening with AI transcription, question generation, and candidate scoring.



GDPR Compliance Toolkit

Candidate data deletion workflows, consent management, export requests, and audit logs.



v2.1

Candidate Referral Engine

Referral tracking, incentive management, and social sharing — turning every hire into a sourcing channel.



Mobile-first Candidate App

React Native app for on-the-go job search, CV scan, and application tracking.

OPEN ITEMS & NEAR-TERM

- Email notification system (in progress)
- Elasticsearch full-text search (optional, configured)
- Advanced team collaboration (comments, @mentions on applications)
- API rate limiting per plan tier
- End-to-end test suite (Playwright)

Vision: PEAXIS becomes the OS layer for every hiring team — from sourcing to onboarding. The AI brain gets smarter with every hire, the candidate experience becomes the competitive advantage, and the data moat makes PEAXIS the default choice for growing companies.

Thank you

Questions welcome

PROBLEM SOLVED	AI-powered hiring with full explainability + end-to-end platform — addressing SME recruitment inefficiencies in Tunisia & globally
TECH STACK	Next.js · NestJS · FastAPI · PostgreSQL + pgvector · Redis · BullMQ · Gemini · Stripe
ARCHITECTURE	4 microservices · Multi-tenant SaaS · Queue-based AI pipeline · Cache-first strategy
DELIVERED	14 API modules · 60+ endpoints · 6 AI features · 4 Docker services · Production-ready
INTERNSHIP CONTEXT	Developed independently during 6-month internship at Prospector · SaaS & AI architecture principles transferred from sales to HR tech domain

BIBANI Mohamed Habib Allah

École Supérieure Privée d'Ingénierie et de Technologie — ESPRIT

Internship at Prospector · AI Sales SaaS → AI Hiring Platform
